

DASHBOARD DOCUMENTATION – TELCO CUSTOMER CHURN

1. Dashboard Overview

This dashboard provides a comprehensive view of customer churn, revenue impact, and service usage. It is designed to support executives and managers in identifying churn drivers, monitoring customer segments, and guiding strategic retention initiatives.

The dashboard is interactive, with slicers, drill-through pages, and bookmarks to enable flexible exploration.

2. Dataset Description

- **Source:** Telco Customer Churn dataset (Excel file).
- **Key Fields:** Customer ID, Gender, Senior Citizen, Partner, Dependents, Tenure, Phone Service, Internet Service, Online Security, Tech Support, Device Protection, Streaming Services, Contract Type, Payment Method, Monthly Charges, Total Charges, Churn.
- **Granularity:** Customer-level data.
- **Timeframe:** Tenure measured in months; revenue tracked via monthly and total charges.

3. Data Model

- **Fact Table:** Customer records with churn status, charges, and tenure.
- **Dimension Tables:**
 - Demographics (Gender, Senior Citizen, Partner, Dependents)
 - Services (Phone, Internet, Security, Tech Support, Device Protection, Streaming)
 - Contracts & Payments (Contract Type, Payment Method)
- **Relationships:** Customer ID as primary key linking dimensions to fact table.

4. DAX Measures

- **Total Customers** = COUNTROWS ('WA_Fn-UseC_-Telco-Customer-Churn (2)')
- **Churned Customers** = CALCULATE (COUNTROWS ('WA_Fn-UseC_-Telco-Customer-Churn (2)'), 'WA_Fn-UseC_-Telco-Customer-Churn (2)' [Churn] = "Yes")

- **Active Customers** = CALCULATE (COUNTROWS ('WA_Fn-UseC_-Telco-Customer-Churn (2)'), 'WA_Fn-UseC_-Telco-Customer-Churn (2)' [Churn] = "No")
- **Churn Rate** = DIVIDE ([Churned Customers], [Total Customers], 0)
- **Retention Rate** = 1 – [Churn Rate]
- **Average Tenure** = AVERAGE('WA_Fn-UseC_-Telco-Customer-Churn (2)' [Tenure])
- **Average Monthly Charges** = AVERAGE('WA_Fn-UseC_-Telco-Customer-Churn (2)' [MonthlyCharges])
- **Revenue Lost to Churn** = CALCULATE(SUM('WA_Fn-UseC_-Telco-Customer-Churn (2)' [MonthlyCharges]), 'WA_Fn-UseC_-Telco-Customer-Churn (2)' [Churn] = "Yes")
- **Total Charges** = SUM('WA_Fn-UseC_-Telco-Customer-Churn (2)' [TotalCharges])
- **Customer Lifetime Value (Estimated)** = [Average Monthly Charges] × [Average Tenure]

5. Dashboard Pages

1. **Executive Overview** – KPIs for churn, retention, revenue, and tenure.
2. **Customer Insights** – Demographics and churn breakdown (gender, senior citizens, dependents, partner status).
3. **Service & Churn** – Churn analysis by services (internet, phone, security, tech support, streaming, multiple lines).
4. **Revenue Dashboard** – Revenue by contract type, lost revenue, monthly/total charges, customer segments.

6. Visual Descriptions

- **KPI Cards:** Total Customers, Churned Customers, Churn Rate, Retention Rate, Revenue Lost.
- **Bar/Column Charts:** Churn by demographics, contract type, payment method.
- **Pie/Donut Charts:** Service adoption vs churn.

- **Line Charts:** Tenure vs churn rate.
- **Histograms:** Distribution of monthly charges.

7. Business Assumptions

- Customer lifetime value is estimated using average tenure $\times$ average monthly charges.
- Revenue lost is calculated based on monthly charges of churned customers, not total charges.
- Churn is binary (Yes/No) and represents complete customer exit.
- Tenure is treated as a proxy for customer loyalty.

8. Limitations

- Dataset does not include external market factors (competitor pricing, promotions).
- Revenue lost measure assumes churned customers stop paying immediately.
- Customer lifetime value is estimated and may not reflect actual retention behavior.
- No temporal churn trend (dataset is static, not time-series).
- Payment method churn analysis may be influenced by demographic overlaps not fully captured.